CIG-STF: Change Information Guided Spatiotemporal Fusion for Remote Sensing Images

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***Abstract*— Spatiotemporal fusion has been attracting increasing attention in remote sensing applications, such as environmental monitoring and land cover change detection, due to its excellent ability to obtain high spatial and temporal resolution images. The land cover change has always been a great challenge in spatiotemporal fusion. Although most spatiotemporal fusion methods have demonstrated satisfactory performance in addressing phenological changes, the performance in terms of abrupt land cover type changes, such as floods or mudslides, falls short. To alleviate this issue, we propose a change information guided spatiotemporal fusion (CIG-STF) method. The proposed CIG-STF integrates change detection and spatiotemporal fusion in a unified framework, by taking advantage of change detection in capturing land cover changes to assist spatiotemporal fusion. Specifically, the CIG-STF comprises three modules: multiscale dilated feature extractor module (MDFE), spatiotemporal fusion-change detection integrated module (STF-CD), and reconstruction module. The MDFE employs multiscale dilated convolutions to comprehensively extract features, to prevent crucial information loss by increasing the convolutional receptive field. In the STF-CD, a change detection module on attention strategy is integrated into the spatiotemporal fusion task, by excavating land cover changes to further enhance the fusion performance. In addition, we design a dynamic decay loss function to further leverage change information, ensuring the accuracy of both change information and prediction results. The experiments were verified on the publicly available LGC and Daxing datasets with manual change labels. The experimental results demonstrate the superior performance of the proposed CIG-STF in both phenological variations and land cover type changes.**

***Index Terms*— Change detection, deep learning (DL), land cover changes, remote sensing, spatiotemporal fusion.**

I. Introduction

REMOTE sensing images with high spatial and temporal resolutions facilitate numerous applications, such as environmental monitoring, disaster management, and land cover change detection [1]. However, owing to technical and financial constraints [2], acquiring images with both high temporal and high spatial resolutions from a single satellite sensor remains a significant challenge, which, to some extent, hinders the broader applications of remote sensing. Therefore, it is imperative to employ spatiotemporal fusion that integrates the diverse advantages of multisource remote sensing images to obtain highly spatial and temporal resolution remote sensing images. For instance, the Landsat-8 satellite offers a spatial resolution of 30 m with a revisit period of 15 days [3], while the MODIS satellite provides a spatial resolution of 250/500/1000 m with a revisit period of just one day [4]. The spatiotemporal fusion technique combines the respective strengths of multisource remote sensing satellites, by extracting the temporal change information from MODIS and the detailed ground textures from Landsat, to obtain the high spatial and temporal resolution fused images. For the sake of simplicity below, we refer to images with high spatial resolution and long revisit period as fine images, and images with low spatial resolution and short revisit period as coarse images.

Spatiotemporal fusion has garnered significant attention within the academic community. Currently, the existing spatiotemporal fusion methods can be divided into traditional methods and the recently proposed deep learning (DL)-based methods. Traditional methods include weight-based methods, spectral unmixing-based methods, and model optimization-based methods. DL plays a vital role in spatiotemporal fusion because of its powerful data processing capabilities and learning capabilities [5], [6]. Among them, the framework based on convolutional neural network (CNN) [7], recurrent neural network (RNN) [8], generative adversarial network (GAN) [9], and transformer-based network (TF) [10] is the mainstream frameworks in the recent spatiotemporal fusion.

Despite the proposal of various spatiotemporal fusion methods that have yielded favorable results, they still encounter certain challenges. Currently, most spatiotemporal fusion methods have demonstrated satisfactory outcomes in addressing phenological changes, but the performance in terms of land cover type changes falls short. Although some DL-based spatiotemporal methods have noticed this issue and made some improvements, they tend to excel primarily in minor

Manuscript received 18 March 2024; revised 22 April 2024; accepted 3 June 2024. Date of publication 18 June 2024; date of current version 21 June 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42171326 and Grant 42371331, in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LR23D010001 and Grant LY22F010014, and in part by the Ningbo Natural Science Foundation under Grant 2022J076. *(Corresponding authors: Xiangchao Meng; Randi Fu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3412118

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or inconspicuous land cover changes, while their performance is limited in areas exhibiting substantial land cover changes, such as floods or mudslides. On the one hand, this might be attributed to the innately complex relationships caused by the differences in sensor hardware, and imaging systems, especially in land cover changes, for which the spectral and spatial information undergo drastic changes, making the relationship more complex. On the other hand, most spatiotemporal fusion methods have given limited consideration to land cover changes in the design of the fusion model.

To tackle the above problems, we proposed a change information guided spatiotemporal fusion (CIG-STF) method. This method incorporates a change detection task to deeply mine land cover changes between dual-temporal images. The change detection task possesses a robust capability to identify the changes between multitemporal images and effectively determines "where is changed." Leveraging this, we employ change detection to assist the spatiotemporal task to obtain fine images of the target moment. The contributions are given as follows.

1) We proposed a CIG-STF method. The proposed method integrates change detection and spatiotemporal fusion in a unified framework, by taking advantage of the excellent ability of change detection in capturing land cover changes to assist spatiotemporal fusion, especially for images with land cover type changes. To the best of our knowledge, the CIG-STF is the first attempt to take advantage of change detection to assist in promoting spatiotemporal fusion.
2) In the proposed method, we design an interesting spatiotemporal fusion-change detection integrated module (STF-CD), by ingeniously collaborating spatiotemporal fusion and change detection, and the change detection is constructed on attention strategy to capture variations between dual-temporal images, to guide and enhance the spatiotemporal fusion model.

The remainder of this article is organized as follows. In Section II, we briefly outline spatiotemporal fusion, change detection, and their relations. Section III presents the detailed framework of our proposed CIG-STF model. Section IV provides the experimental details and comprehensive analysis. Finally, we present the conclusion of our work in Section V.

II. Related Work

A. Spatiotemporal Fusion

Most existing spatiotemporal fusion algorithms can be roughly divided into four categories: weight-based, spectral unmixing-based, model optimization-based, and DL-based methods.

Weight-based methods leverage the concept that changes in land surface reflectance can be predicted by a linear model that aggregates the temporal–spatial image pixels and applies a weighted-sum strategy. A typical weight-based spatial and temporal adaptive reflectance fusion model (STARFM) [11] is founded on a linear assumption regarding changes in reflectance. It establishes a straightforward approximation model for delineating the connection between fine and coarse images. However, it is important to note that a pixel may contain more than one land cover type. In situations where pixels encompass complex land cover types, the effectiveness of STARFM may be diminished. Consequently, several other weight-based algorithms, building upon the theoretical foundations of STARFM, have been proposed to address these issues. The enhanced STARFM (ESTARFM) [12] algorithm introduces differences between mixed pixels and pure pixels to modify the weights of STARFM. The modified ESTARFM (mESTARFM) [13] introduces a novel approach for searching similar pixels to enhance the fusion performance. The spatiotemporal adaptive algorithm for mapping reflectance change (STAARCH) [14] takes into consideration the differences between multiple coarse images, and it demonstrates significant capability in predicting local spectral changes. Nonetheless, due to the linear assumptions among multisource and multitemporal images, their fusion effectiveness is somewhat limited in cases of significant land cover type variations.

Spectral unmixing-based methods are founded on the principle of spectral unmixing theory, assuming that a pixel can be divided into abundance and endmember values. The multisensor multiresolution technology (MMT) [15] derives the predicted fine images by utilizing the unmixing results from the target coarse images and the end members of adjacent fine images. The spatiotemporal data fusion method (STDFA) [16] method assumes that the land cover type remains unchanged during the prediction time range. Compared with the STDFA algorithm, the enhanced spatiotemporal data fusion model (ESTDFM) [17] enhances the end-member classification and sets a movable window and time weight, which makes a more accurate fused performance. The flexible spatiotemporal data fusion (FSDAF) [18] comprehensively integrates the benefits of unmixing-based and weight-based methods and also achieves good results. Although the performance of these methods has been improved, the performance in areas with complex land cover types still needs to be improved.

Model optimization-based methods aim to establish the relationship between the reference and target images to predict the fine-resolution images. These methods encompass Bayesian- and sparse representation-based methods. The Bayesian methods [19], [20], [21] generally offer good interpretability. However, their fusion efficiency is relatively lower due to the requirement for iterative solutions. The core principle of methods based on sparse representation or dictionary learning is found in compressed sensing theory to establish a spatial and temporal relationship between the fused images and the observations, such as sparse representation-based spatiotemporal reflection fusion model (SPSTFM) [22]. Inspired by the SPSTFM, many improved methods have been proposed, such as error bound regularized sparse coding (EBSPTM) [23]. Nevertheless, these methods have some drawbacks. First, it is challenging to model the nonlinear relationships between different spatial and temporal resolution images. Second, the operational speed of traditional methods needs improvement, and there is room for enhancement in fusion accuracy.

In recent years, DL has sparked a research trend in the academic community due to its outstanding performance in the field of image data processing. The spatiotemporal fusion

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based on a deep CNN (STFDCNN) [24] can be considered as the pioneering algorithm, which utilizes a super-resolution CNN to produce high-resolution images at the predicted time. Tan et al. [25] proposed a deep convolution model for spatiotemporal fusion (DCSTFN) that directly integrates the difference images (DIs) between coarse images into reference fine images to obtain the target fine images. Based on DCSTFN, the enhanced deep convolutional spatiotemporal fusion network (EDCSTFN) [26] was proposed to further improve the performance. Li et al. [27] introduced a DMNet, by combining dilation convolution and multiscale mechanisms. This approach primarily aims to prevent information loss and redundancy during the convolution process. Liu et al. [28] introduced a two-stream CNN (StfNet) that takes into account the temporal dependencies within time series, and it compensates for the loss of spatial information in coarse images that occurs with other methods. Meng et al. [29] proposed a pioneering spatial–temporal–spectral collaborative learning network (STFRNN) model, which combines CNNs and RNNs to generate high-quality target fine images. Liu et al. [30] presented a detail-injected spatiotemporal fusion (DISTF) model by exploring the intrinsic relationship between spatiotemporal fusion and spatiospectral fusion. Zhang et al. [31] introduced a spatiotemporal fusion method based on GAN (STFGAN), which alleviates the large difference in spatial and temporal resolution between coarse and fine images. Tan et al. [32] proposed a flexible spatiotemporal fusion model based on conditional GANs (GAN-STFM) that only requires one target time image and a random coarse image at other times, breaking the time limit on reference image selection. Song et al. [33] proposed a new multilevel feature fusion model based on the GAN network (MLFF-GAN) by considering the huge resolution difference between high- and low-resolution images.

B. Change Detection

Change detection task for remote sensing aims to identify changes in land cover types between images captured at the same location during different phases. Based on its robust identification capabilities, it has wide applications in Earth surface exploration, such as infrastructure management, ecological monitoring, and urban expansion. Currently, the proposed change detection methods can be divided into traditional methods and DL-based methods.

Traditional methods can commonly be divided into pixel-based methods and object-based methods. Pixel-based methods typically begin by comparing every pixel between multitemporal images to calculate their DI and then classify each pixel to generate the change maps. These algorithms include methods based on arithmetic, such as image rationing or image differencing [34], methods based on transformations such as principal component analysis (PCA) [35], [36] or change vector analysis (CVA) [37], [38], and methods based on classification, among other algorithms [39], [40]. However, pixel-based methods primarily focus on changes between individual pixels during different phases, often overlooking significant global image information. In contrast, object-based methods segment images according to characteristics such as image shape, detail information, and spectral changes, allowing for a more comprehensive analysis of image data. Nonetheless, these object-based methods still face a lot of challenges, the misclassification errors resulting from the uncertainty of segmented objects propagating during the different phases of change detection, potentially constraining the overall detection accuracy. In summary, these traditional methods suffer from certain issues, ultimately resulting in unsatisfactory change maps.

DL-based methods can effectively alleviate these problems and achieve remarkable results. CNN-based network possesses formidable feature extraction capabilities, enabling it to extract spatial depth features more effectively, integrate contextual information to capture changing interesting regions, and yield precise change maps. Zhang and Shi [41] introduced a change detection method based on a feature difference CNN (FDCNN) that utilizes CNN to extract deep features from multitemporal remote sensing images and incorporates transfer learning to produce difference maps. Fang et al. [42] designed the concept of a densely connected Siamese network for change detection (SNUNet-CD), which can improve the utilization of shallow features and address the issue of missing position information in deep features. Immediately afterward, GAN-based methods have seen extensive utilization in remote sensing image change detection. He et al. [43] introduced a change detection network that utilizes an encoder–decoder as the GAN generator (UNet-GAN-CD), resulting in improved change detection performance. Zhao et al. [44] presented an attention gate GAN-based change detection network (AG-GAAN) designed for multiple change detection, resulting in enhanced model accuracy in detecting multiple changes. Recently, transformer-based methods have also been incorporated into change detection research. Zhang et al. [45] introduced a Siamese U-shaped change detection network based on a transformer (SwinSUNet) to mitigate the limitations of convolutional operations, which primarily focus on local information. Yan and Jiang [46] proposed a hybrid Siamese network that combines CNN and transformer (HSSEN), which effectively enhances and refines the different features.

Furthermore, research attention is also directed toward change detection in heterogeneous remote sensing images. Li et al. [47] proposed a deep transformation network (DTCDN) for change detection in optical and SAR images, aiming to reduce the differences between multimodalities. Liu et al. [48] introduced a domain adaptation-based cross-reconstruction change detection (DAHCD) model that effectively improves credibility during the conversion stage.

C. Relations Between Spatiotemporal Fusion and Change Detection

Spatiotemporal fusion and change detection tasks are essential in their respective domains. Yet, due to the shared factor of "change" in their execution processes, there are both similarities and differences between them that warrant further exploration.

The spatiotemporal fusion task usually involves extracting fine ground texture information from the auxiliary images of

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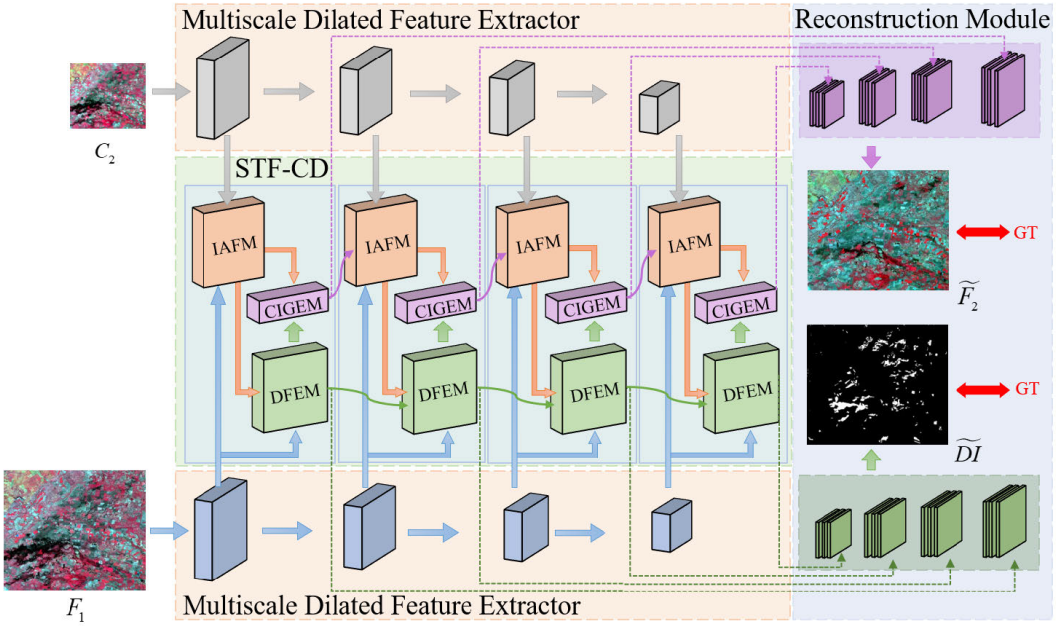


Fig. 1. Overall framework of the CIG-STF model. The CIG-STF consists of a multiscale dilated feature extractor, an STF-CD module, and a reconstruction module. The red arrows represent the loss function and the other arrows represent feature flow.

other time phases and capturing temporal change information between the auxiliary and the target moment, to generate high-resolution images for the target moment. The change detection tasks generally focus on mining the change maps from dual-temporal images, usually represented as a binary map with 0 indicating unchanged regions and a pixel value of 1 showing the changed location. It can be concluded that the difference between the two tasks lies in the fact that spatiotemporal fusion aims to analyze what has "changed" in comparison to the previous moment images, such as floods and landslides, which has practically meaningful insights. In contrast, change detection tasks aim to determine whether a change has occurred at a certain location, essentially obtaining spatial-temporal change position information. Certainly, it should also be noted that the multiclass change detection methods are attracting attention, to simultaneously tell us "where is changed," "what is changed," and "changed to what."

Despite these differences, these two tasks share a commonality; they both involve using images from different temporal moments and analyzing the changes between them. Consequently, the change maps obtained from change detection tasks can be leveraged in the spatiotemporal fusion process. Specifically, the change maps are utilized to assist spatiotemporal fusion tasks in precisely locating areas where changes occurred in the target moment relative to the previous moment. This approach can precisely standardize the spatial extent of the changing regions, ultimately improving the accuracy of generating fine images for the target moment.

III. Methodology

In this article, we proposed a CIG-STF method. The network architecture is illustrated in Fig. 1. Specifically, the proposed model consists of three parts: multiscale dilated feature extractor (MDFE), STF-CD, and reconstruction module. First, we just take the fine image F_1 at time t_1 and the coarse image C_2 at time t_2 as the inputs into the MDFE to achieve the multiscale feature maps. Then, the extracted high-resolution F_1 feature maps and the low-resolution C_2 feature maps are input into the STF-CD model to achieve image fusion and change information extraction. The STF-CD module can be divided into three parts: invertible attention fusion module (IAFM), difference feature extractor module (DFEM), and change information-guided enhancement module (CIGEM). In the initial phase, we utilize the F_1 and C_2 feature maps obtained from MDFE, to generate preliminary prediction results through the IAFM. Subsequently, we design an attention strategy in the DFEM to extract change maps between these results and feature maps from F_1. Finally, we introduce the CIGEM to inject the change maps into the preliminary prediction outcomes, producing an enhanced final spatiotemporal fusion result. Subsequently, we will introduce each component of the proposed STF-CD in detail. The reconstruction module amalgamates the outputs from each layer and obtains the ultimate predicted results and the change maps. Concurrently, we have devised a hybrid loss function to impose constraints on the change maps and the predicted outcomes, aligning them more closely with the ground truth. This dynamic strategy can be more flexible to maintain the authenticity of prediction results and improve model robustness as much as possible.

A. Multiscale Dilated Feature Extractor

Remote sensing images contain abundant spatial and spectral information, and it is hard to extract valuable features through a single convolution operation. The final high-resolution images are generated by combining detailed texture information from fine images and temporal ground change information from coarse images. Therefore, extracting useful features from remote sensing images is key to producing

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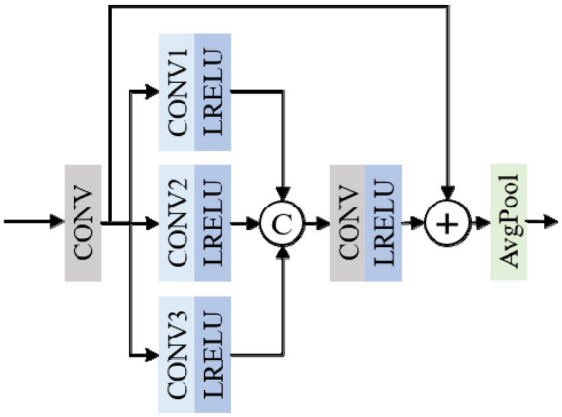


Fig. 2. Framework of the MDFE module.

high-resolution images. Convolutional operations are constrained by the receptive field, and a single convolutional layer just can extract features at a single scale, limiting its capability to capture a diverse range of features. This may result in the omission of significant information and an inability to capture global context. As a result, we designed the MDFE module for extracting multiscale features, as illustrated in Fig. 2. In this module, we introduce dilated convolutions, which effectively increase the receptive field without increasing the number of parameters. Consequently, it can focus extensively on valuable information within the images, preventing information loss and extracting the necessary features. As the network inputs originate from different sensors and contain distinct information, although we adopt the same feature extraction framework, but refrain from weight sharing, this enables the network to extract more representative features from each, facilitating subsequent operations. The process of the proposed MDFE module can be illustrated as

$$F_1^i = \mathrm{Avg}(\mathrm{block}(\mathrm{Concat}(\mathrm{block}_1(\mathrm{Conv}(F_1)), \mathrm{block}_2 \times (\mathrm{Conv}(F_1)), \mathrm{block}_3(\mathrm{Conv}(F_1)))) \oplus \mathrm{Conv}(F_1)) \tag{1}$$

$$C_2^i = \mathrm{Avg}(\mathrm{block}(\mathrm{Concat}(\mathrm{block}_1(\mathrm{Conv}(C_2)), \mathrm{block}_2 \times (\mathrm{Conv}(C_2)), \mathrm{block}_3(\mathrm{Conv}(C_2)))) \oplus \mathrm{Conv}(C_2)) \tag{2}$$

where F_1 and C_2 are the fine images and the coarse images, respectively, and block is the combination of a convolution with kernel 3 and a Leaky ReLU activation function. block_1, block_2, and block_3 are the combinations of the convolution with kernel 3, dilation rates of 1–3, and leaky ReLU, respectively. Concat represents the concatenate operation and Avg represents the pooling operation. F_1^i and C_2^i represent the feature maps after MDFE extraction, where i represents the number of layers.

B. STF-CD Module

The proposed STF-CD module can be considered as a spatiotemporal fusion-change detection integrated process framework, and the construction of the STF-CD module is illustrated in Fig. 3. By utilizing meticulously crafted modules, each designed for specific objectives, and we leverage the extracted change information to effectively guide and enhance the final fusion results. Therefore, the STF-CD module can be subdivided into three distinct submodules as follows.

1) Invertible Attention Fusion Module: To achieve the preliminary result of spatiotemporal fusion, we designed the IAFM and input F_1^i and C_2^i obtained through the above MDFE. The specific structure of the module is illustrated in Fig. 3(a). Landsat and MODIS images encompass intricate spatial–spectral information, resulting in the need for necessary operations to integrate, which is essential for extracting the spatial or spectral information contained within them. The spatial attention module (SAM) operates on the spatial dimension of feature maps, capturing the spatial positions of interest by modeling contextual spatial information. The channel attention module (CAM) works on the channel dimension of feature maps, obtaining crucial spectral information by modeling weighted interactions across channels. Clearly, SAM and CAM can effectively integrate spatial and spectral information from feature maps, respectively, so we use these two modules to extract spatial and spectral information from features.

First, we extract the spectral information from C_2^i using CAM and distill the spatial attention weight from F_1^i through SAM, which represents the spatial location information within F_1^i. Subsequently, we first concatenate F_1^i and C_2^i feature maps and then use the spatial attention weight maps obtained from F_1^i above to modify the concatenated feature, thereby obtaining advantage spatial location information. Following this, the obtained spectral features from C_2^i are fused with the crucial spatial location information, achieving features dominated by F_1^i spatial positions and C_2^i spectral information. To fully exploit the spatial–spectral information in F_1^i and C_2^i, we repeat the above operations in the IAFM, but with an inversion of the dominant entities, resulting in features dominated by C_2^i spatial positions and F_1^i spectral information. Finally, the preliminary features dominated by different entities are added together, resulting in high-resolution prediction features $\widehat{F_2^i}$ that simultaneously possess rich ground texture information and temporal change information. Summing up, the specific operational formulas for this module are given as follows:

$$F^i_{F_1^i \to C_2^i} = \mathrm{Concat}\big(\mathrm{SAM}\big(F_1^i\big) \otimes \mathrm{Concat}\big(F_1^i, C_2^i\big), \mathrm{CAM}\big(C_2^i\big) \otimes C_2^i\big) \tag{3}$$

$$F^i_{C_2^i \to F_1^i} = \mathrm{Concat}\big(\mathrm{SAM}\big(C_2^i\big) \otimes \mathrm{Concat}\big(F_1^i, C_2^i\big), \mathrm{CAM}\big(F_1^i\big) \otimes F_1^i\big) \tag{4}$$

$$\widehat{F_2^i} = F^i_{F_1^i \to C_2^i} \oplus F^i_{C_2^i \to F_1^i} \tag{5}$$

where SAM and CAM are the SAM and CAM, respectively; $\otimes$ is the element-wise multiplication; $\oplus$ is the element-wise summation; $F^i_{F_1^i \to C_2^i}$ represents the features dominated by F_1^i spatial location information and C_2^i spectral information; $F^i_{C_2^i \to F_1^i}$ represents the features dominated by C_2^i spatial location information and F_1^i spectral information; and $\widehat{F_2^i}$ represents the preliminary prediction results.

2) Difference Feature Extractor Module: Given that the above IAFM has obtained the preliminary prediction results

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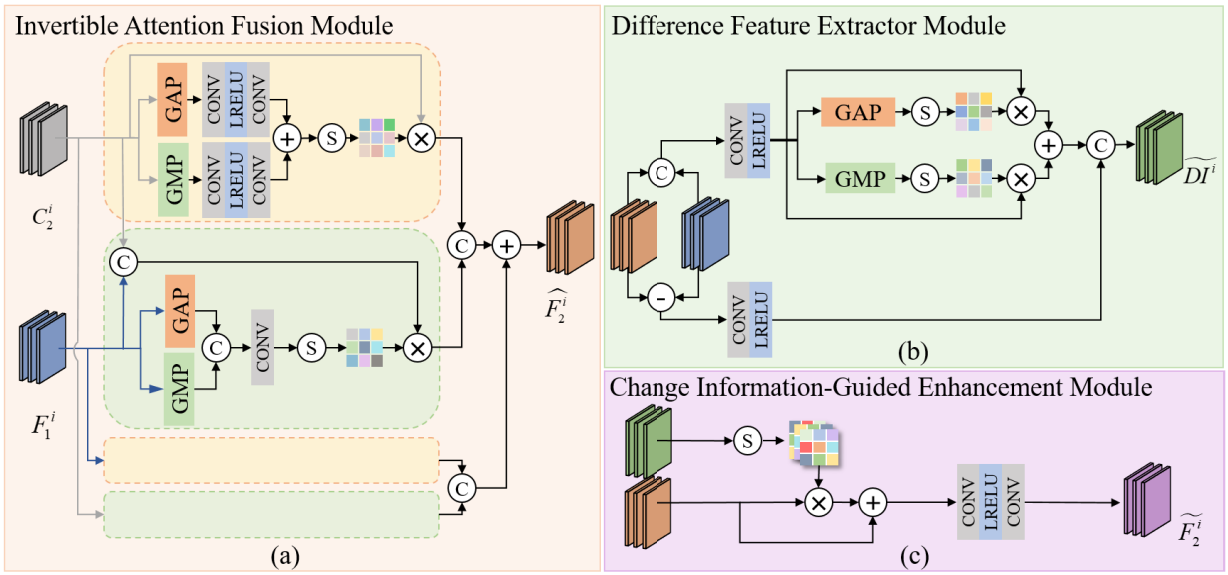


Fig. 3. Framework of STF-CD module. (a) Invertible attention fusion module. (b) Difference feature extractor module. (c) Change information-guided enhancement module.

$\widehat{F_2^i}$, it already has the temporal changes and ground texture information for the target moment. Thus, we design the DFEM, which leverages the preliminary fused features $\widehat{F_2^i}$ and the features of F_1^i, to learn the mapping of changes between them. The module framework is illustrated in Fig. 3(b). Specifically, we first concatenate the dual-temporal features to obtain comprehensive feature maps. We believe that the prominent parts of the fused features belong to the changing area. Therefore, implementing the attention mechanism, we utilize parallel branches with global average pooling (GAP) and global max pooling (GMP) to extract global and local estimated valuable weights σ_1 and σ_2, respectively. Next, multiply with the fused features and add these feature maps to obtain the changing area of interest in the dual-temporal feature maps. Finally, we merge these features with the difference feature maps obtained by subtraction to achieve the ultimate change map results. The computational process is outlined as follows:

$$\widehat{\mathrm{DI}^i_{\mathrm{concat}}} = \mathrm{block}\left(\mathrm{Concat}\left(\widehat{F_2^i}, F_1^i\right)\right) \tag{6}$$

$$\widehat{\mathrm{DI}^i_{\mathrm{sub}}} = \mathrm{block}\left(\widehat{F_2^i} - F_1^i\right) \tag{7}$$

$$\sigma_1 = \mathrm{Sig}\left(\mathrm{GMP}\left(\widehat{\mathrm{DI}^i_{\mathrm{concat}}}\right)\right) \tag{8}$$

$$\sigma_2 = \mathrm{Sig}\left(\mathrm{GAP}\left(\widehat{\mathrm{DI}^i_{\mathrm{concat}}}\right)\right) \tag{9}$$

$$\widetilde{\mathrm{DI}^i} = \mathrm{block}\left(\mathrm{Concat}\left(\sigma_1 \otimes \widehat{\mathrm{DI}^i_{\mathrm{concat}}} \oplus \sigma_2 \widehat{\mathrm{DI}^i_{\mathrm{concat}}}, \widehat{\mathrm{DI}^i_{\mathrm{sub}}}\right)\right) \tag{10}$$

where $\widehat{\mathrm{DI}^i_{\mathrm{concat}}}$ and $\widehat{\mathrm{DI}^i_{\mathrm{sub}}}$ are the combination of the dual-temporal features and the difference features of the dual-temporal features, respectively; Sig is the sigmoid function; σ_1 and σ_2 are the weights of the combined features; and $\widetilde{\mathrm{DI}^i}$ represents the final change maps.

3) Change Information-Guided Enhancement Module: After applying the aforementioned IAFM and DFEM, we obtain preliminary prediction results $\widehat{F_2^i}$ and change maps $\widetilde{\mathrm{DI}^i}$. At this point, it is essential to explore the facilitating role of change detection results on spatiotemporal fusion. Therefore, we introduce the CIGEM to enhance the generation of superior final prediction results. The specific framework is illustrated in Fig. 3(c). As we all know, change detection tasks aim to identify whether there have been alterations in land cover and typically result in a binary map where 1 signifies somewhere was changed and 0 denotes somewhere was not changed. The generated change map results $\widetilde{\mathrm{DI}^i}$ by the aforementioned DFEM are weight maps, so we employ the sigmoid function to transform it into weights ranging from 0 to 1. In these weighting maps, a higher value indicates a larger range of change, and a smaller value indicates a smaller range of change. Therefore, the multiplication operation can enhance the learning of the variable region and the suppression of the invariant region in the preliminary fusion image. Finally, the initial fusion image is added to prevent the loss of details, and the final fusion image $\widetilde{F_2^i}$ is obtained through the integration of convolution. The specific calculation process is given as follows:

$$\widetilde{F_2^i} = \mathrm{Conv}_{3\times 3}\left(\mathrm{block}\left(\mathrm{Sig}\left(\widetilde{\mathrm{DI}^i}\right) \otimes \widehat{F_2^i} \oplus \widehat{F_2^i}\right)\right). \tag{11}$$

To safeguard against the loss of low-level information and alleviate the vanishing gradient problem during the hierarchical training process, we downsample the fusion results and change maps obtained at each layer and then employ a skip connection to augment the connections between each level, which is helpful for model learning global feature, thereby improving the performance of the model. It can be concluded that

$$\widetilde{F_2^{i+1}} = \widetilde{F_2^{i+1}} + \widetilde{F_2^i} \downarrow \tag{12}$$

$$\widetilde{\mathrm{DI}^{i+1}} = \widetilde{\mathrm{DI}^{i+1}} + \widetilde{\mathrm{DI}^i} \downarrow \tag{13}$$

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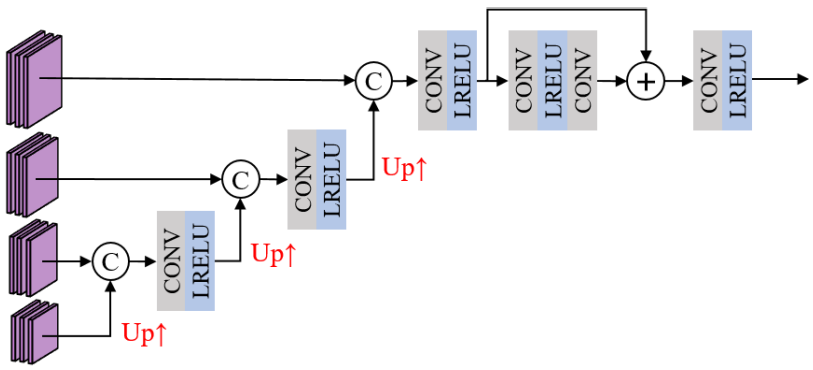


Fig. 4. Framework of the reconstruction module.

where $\downarrow$ represents the downsampling operation and $\widetilde{F_2^{i+1}}$ and $\widetilde{\mathrm{DI}^{i+1}}$ represent the features from the next layer.

C. Reconstruction Module and Loss Function

1) Reconstruction Module: After employing different scale STF-CD modules, we obtained the predicted fusion results and the predicted change map outcomes at various scales. Subsequently, these results were reconstructed individually through a dedicated reconstruction module, yielding the final target time prediction results and change maps for the dual-temporal images. In addition, we contend that relying solely on simple upsampling and concatenation operations has a limited capacity to recover fine pixels and may lead to the loss of crucial information. Consequently, we introduced a combination of convolution operation with kernel size 3 plus a skip connection after upsampling reconstruction. This design aims to enhance the preservation of valuable information in the original data. The specific structure is shown in Fig. 4. The specific calculation process is given as follows:

$$\widetilde{F_2'} = \sum_{i=1}^{3} \mathrm{block}\left(\mathrm{Concat}\left(\widetilde{F_2^{i+1}} \uparrow, \widetilde{F_2^{i}}\right)\right) \tag{14}$$

$$\widetilde{F_2} = \mathrm{block}\left(\mathrm{Conv}_{3\times 3}\left(\mathrm{block}\left(\widetilde{F_2'}\right)\right) + \widetilde{F_2'}\right). \tag{15}$$

2) Loss Function: It plays a crucial role in DL, and in this task, we use change detection results to enhance spatiotemporal fusion outcomes. Therefore, we design a dynamic decay loss function to obtain high-precision fine images at the target moment. First, we utilize the mean squared error (MSE) and multiscale structural similarity index (MS-SSIM) [49] to calculate the feature loss between the spatiotemporal fusion results and the ground truth. This aids in better preserving spatial texture details in the fused images and ensures a satisfactory visual outcome by maintaining feature consistency and structural similarity with the ground truth. $\mathcal{L}_{\mathrm{feature}}$ is denoted as in (16) and $\mathcal{L}_{\mathrm{structure}}$ is denoted as in (17)

$$\mathcal{L}_{\mathrm{feature}} = \frac{1}{N}\sum_{i=1}^{N}\left[\widetilde{F_2} - \mathrm{GT}\right]^2 \tag{16}$$

$$\mathcal{L}_{\mathrm{structure}} = \mathrm{I} - \left[l_j(\widetilde{F_2}, \mathrm{GT})\right]^{V_k} \cdot \prod_{j=1}^{K}\left[c_j(\widetilde{F_2}, \mathrm{GT})\right]^{P_j}\left[s_j(\widetilde{F_2}, \mathrm{GT})\right]^{\mu_j} \tag{17}$$

where $\widetilde{F_2}$ GT are the prediction images and ground truth images, respectively, and N is the pixel number of the images. In (15), I is the tensor of ones; K represents the number of the highest scale; and l_j, c_j, and s_j are the luminance, the contrast comparison, and the structure comparison at the jth scale of the images. v_k, ρ_j, and v_j are the corresponding weight coefficients.

Subsequently, we incorporate the cross-entropy loss to refine the precision of change maps, which can be denoted as (18). The heightened accuracy of the change maps will correspondingly enhance the accuracy of the predicted fusion results. In addition, to better leverage the auxiliary role of the change maps, we have devised a weight decay strategy for training CIG-STF. This strategy facilitates a shift in the network's focus from the auxiliary task to the primary task. At the beginning of the network training, we focus on the training of the change maps because it can make the preliminary prediction results more accurate and ensure better results in the subsequent training process. As the network continues to deepen, we gradually shift the focus toward the accuracy of the prediction results, thereby better constraining the network to generate more precise predictions and improving network performance. The weight decay parameters can be illustrated as (19)

$$\mathcal{L}_{\mathrm{ce}} = \left[-(\mathrm{GT}\log(\widetilde{F_2}) + (\mathbf{I} - \mathrm{GT})\log(\mathbf{I} - \widetilde{F_2}))\right] \tag{18}$$

$$\gamma = A\cos\left(\frac{e\pi}{2E}\right) \tag{19}$$

where A is the attenuation coefficient and e and E represent the current training epoch and the total epochs of training, respectively. Therefore, the total loss function of the network can be expressed as follows:

$$\mathcal{L}_{\mathrm{total}} = \alpha \cdot \mathcal{L}_{\mathrm{feature}} + \beta \cdot \mathcal{L}_{\mathrm{structure}} + \gamma \cdot \mathcal{L}_{\mathrm{ce}} \tag{20}$$

where α and β are the weight coefficients of $\mathcal{L}_{\mathrm{feature}}$ and $\mathcal{L}_{\mathrm{structure}}$, respectively. We set $\alpha = \beta = 1$ in this experiment.

IV. Experiments

As of now, no existing datasets in the public domain have introduced spatiotemporal fusion datasets with change labels.

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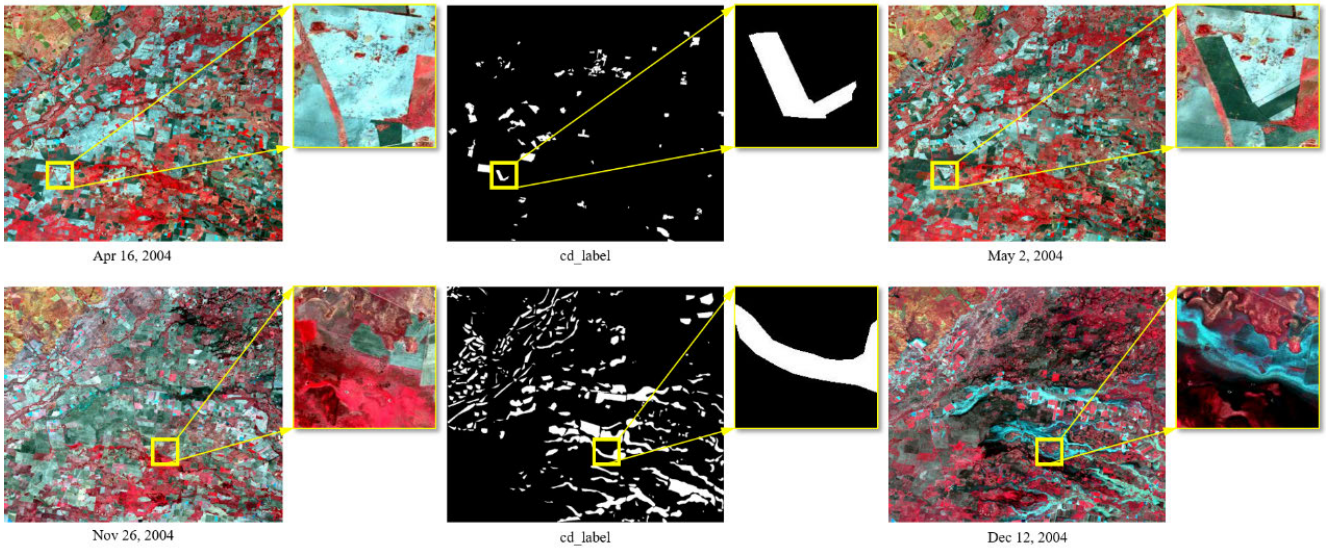


Fig. 5. Example images of the LGC dataset.

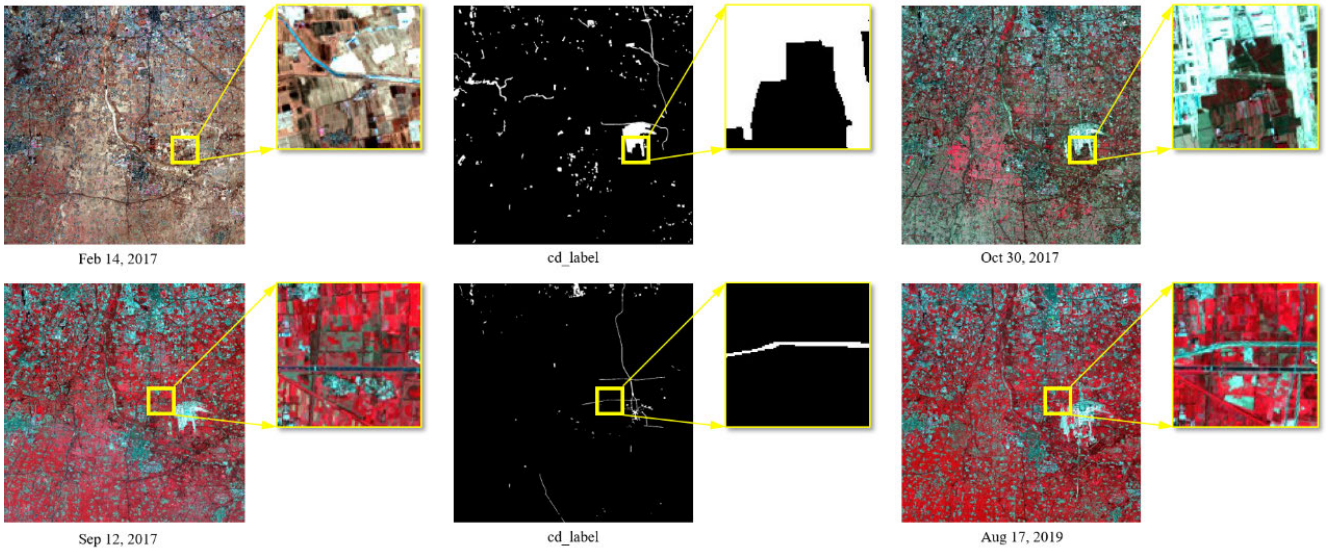


Fig. 6. Example images of the Daxing dataset.

To validate CIG-STF, we took the initiative to annotate change labels on two publicly available spatiotemporal fusion datasets. In addition, we conducted ablation experiments to validate the efficacy of our proposed DFEM module, GIGEM module, and the dynamic decay loss function.

A. Datasets and Training Details

In order to verify the effectiveness of the proposed CIG-STF model, we selected two public spatiotemporal fusion datasets with huge land coverage changes to manually annotate change labels. The first dataset was captured on the lower Gwydir catchment (LGC) areas [50], with the primary feature being land cover changes between 2004 and 2005. This dataset comprises 14 pairs of MODIS-Landsat images, each with a spatial dimension of 3200 × 2720 pixels and spanning across six spectral bands. We select two images at adjacent moments in this dataset as a pair and employ GeoLabel software for annotating land cover changes. The second dataset was captured in the Daxing district located in the south of Beijing city (Daxing) [51], with the primary feature being land cover changes between 2013 and 2019. This dataset comprises 29 pairs of cloud-free MODIS-Landsat images, each with a spatial dimension of 1640 × 1640 pixels and spanning across six spectral bands. Given that the period of this dataset exceeds five years and includes numerous image pairs, there are not sufficient land cover changes observed in two images captured at adjacent moments. To address this, we selected 24 pairs of long-term images from the time-series dataset and utilized GeoLabel software for annotating land cover changes. Ultimately, we derived 11 pairs of Landsat–MODIS image pairs with change labels on the LGC dataset, and 24 pairs

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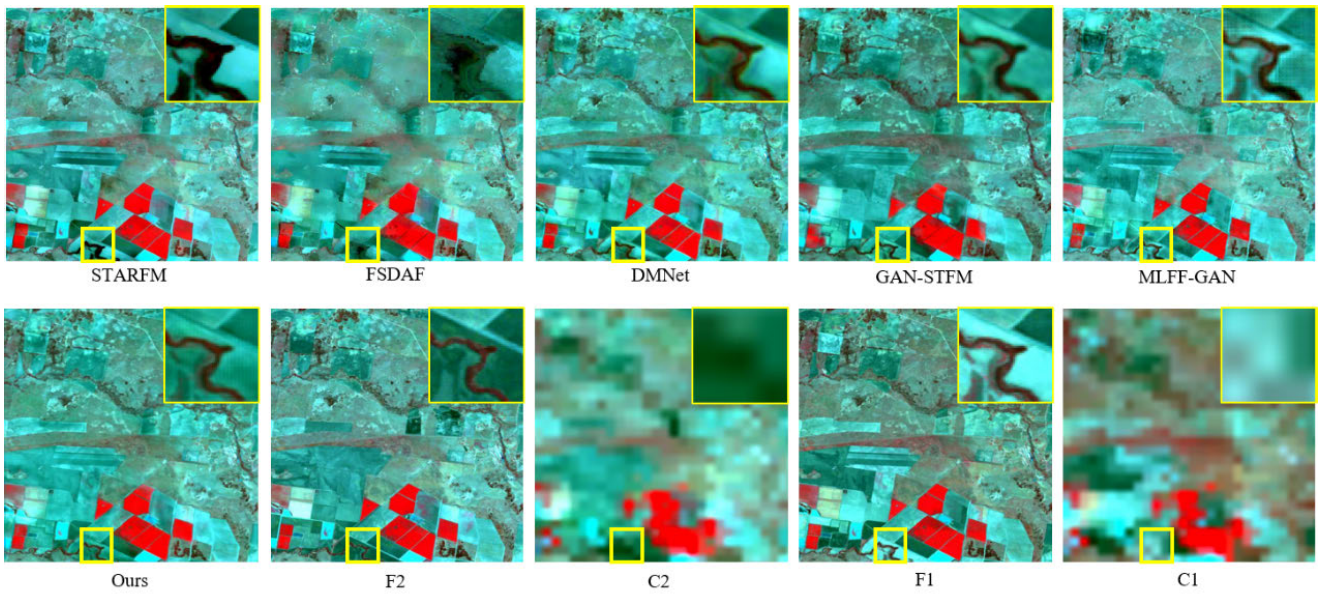


Fig. 7. Fusion results for images with phenological changes in the LGC dataset. Among them, F1 and C1 are the fine and coarse images, respectively; at the auxiliary moment February 14, 2005, C2 is the coarse image at the target moment March 2, 2005; and F2 is the ground truth.

of Landsat–MODIS image pairs with change labels on the Daxing dataset. Figs. 5 and 6 show the example images of the proposed datasets on the LGC and Daxing datasets, respectively.

Each new dataset with change labels is divided into training and testing sets. For the LGC dataset, the images with noticeable ground changes and images with distinct phenological variations (i.e., August 6, December 12, 2004, and March 2, 2005) were selected as the testing datasets, and the remaining images were utilized for the training datasets. For the Daxing dataset, we employed a similar selection method as used for the LGC dataset. Images featuring evident changes in ground objects and phenological variations were designated as the testing datasets (i.e., September 4, 2014–February 14, 2016; August 22, 2015–February 14, 2016; August 22, 2015–December 14, 2016; May 18, 2015–September 12, 2017; August 22, 2015–September 12, 2017; and September 12, 2017–August 17, 2019), while the remaining image pairs were utilized for the training datasets. Due to computational memory constraints, we opt to crop images in the training set into small blocks of 64 × 64 pixels for training purposes. Meanwhile, for the testing set, images were cropped into small blocks of 512 × 512 pixels to enable a more intuitive comparison. The initial learning rate for the training process is set to $2e^{-4}$, the batch size is configured as 16, and the epoch is set to 100 in the experiments.

B. Comparison Methods and Evaluation Indicators

To validate the effectiveness of the proposed CIG-STF model, we chose to compare it with five commonly used spatiotemporal fusion methods. This includes two traditional methods (i.e., STARFM [11] and FSDAF [18]) and three DL-based spatiotemporal fusion methods (i.e., DMNet [27], GAN-STFM [32], and MLFF-GAN [33]).

In the quantitative evaluation, we employed four commonly used metrics. The root-mean-square error (RMSE) measures the difference between predicted and actual images, with lower values indicating higher accuracy. Spectral angle mapper (SAM) [52] assesses spectral similarity between images by measuring the angle between two spectral vectors, with an ideal value of 0. Structural similarity index (SSIM) [53] evaluates the structural similarity between true and predicted values, considering brightness, contrast, and structure; higher SSIM values indicate greater spatial structural similarity. The correlation coefficient (CC) quantifies the linear relationship between two variables, with larger CC values indicating greater similarity between predicted and actual images.

C. Experimental Results and Analysis

1) Experiments for Phenological Changes: Although the change labels of our dataset are annotated based on land cover changes, they are still helpful for the recovery of phenological changes. To prove that our proposed model has significant advantages in recovering phenological changes, we tested images with obvious phenological changes. Figs. 7 and 8 show the qualitative experimental results obtained by STARFM, FSDAF, DMNet, GAN-STFM, MLFF-GAN, and our proposed method on March 2, 2005 and December 14, 2016, respectively. More specifically, since the small gap in phenological changes within the LGC dataset, each comparison method exhibits satisfactory results, as shown in Fig. 7. Notably, the STARFM displays significant distortion at the object's edge (the magnified area), while FSDAF exhibits spatial distortion details in extensive areas that are challenging to reconstruct (the magnified area). Several DL-based methods, DMNet, GAN-STFM, and MLFF-GAN, demonstrate similar great effects, but they still manifest certain spectral distortions and details lost. In contrast, our proposed CIG-STF method exhibits the closest resemblance

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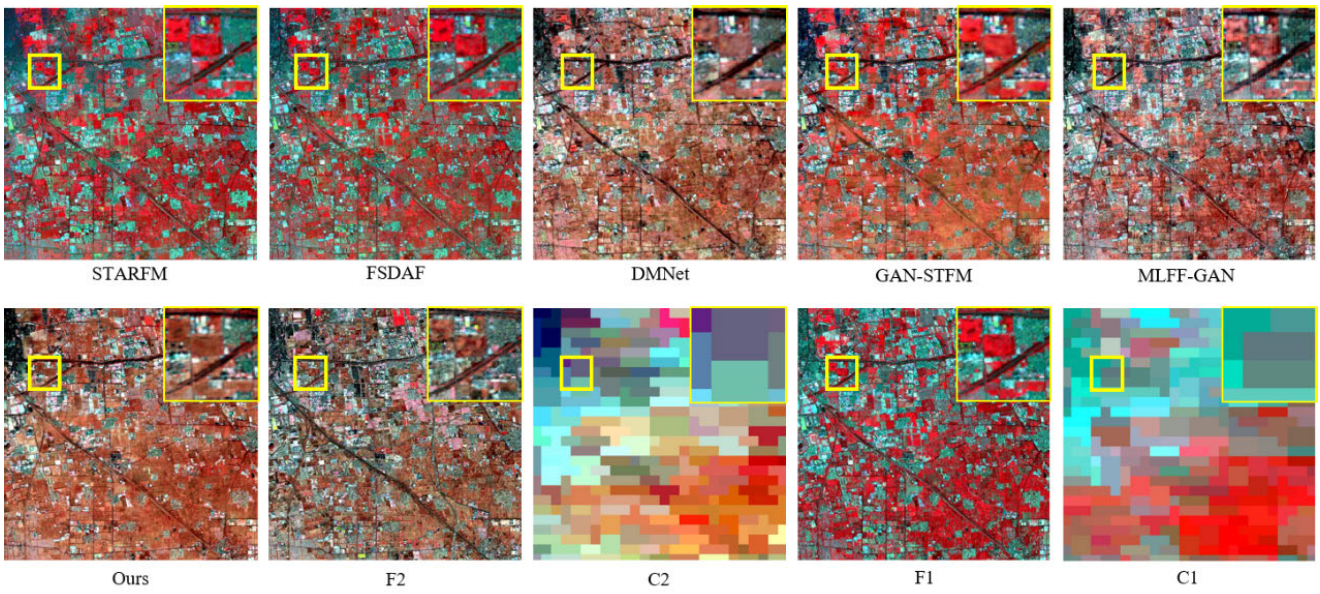


Fig. 8. Fusion results for images with phenological changes in the Daxing dataset. Among them, F1 and C1 are the fine and coarse images, respectively; at the auxiliary moment August 22, 2015, C2 is the coarse image at the target moment December 14, 2016; and F2 is the ground truth.

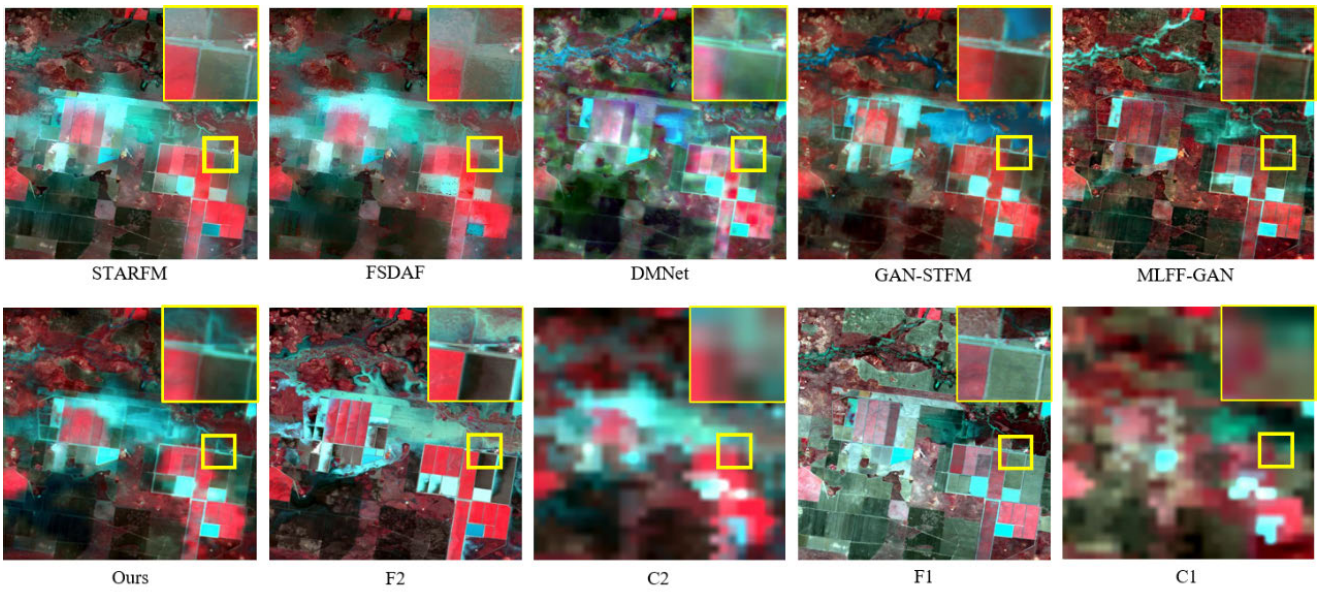


Fig. 9. Fusion results for images with land cover changes in the LGC dataset. Among them, F1 and C1 are the fine and coarse images, respectively; at the auxiliary moment November 26, 2004, C2 is the coarse image at the target moment December 12, 2004; and F2 is the ground truth.

to the image at the target time in terms of spatial detail fidelity and spectral characterization. This means that our method can achieve a high-fidelity recovery of phenological changes.

Given the interval between each pair of data in our Daxing dataset with change labels spanning several years, resulting in significant phenology changes, the performance requirements of the network become more stringent. As illustrated in Fig. 8, the STARFM and FSDAF struggle to recover spectral characteristics that are similar to the predicted image at the target moment. In comparison, the DL methods (DMNet, GAN-STFM, and MLFF-GAN) have achieved relatively good performance, but there are still errors in spectral recovery at the detailed level, and it cannot accurately depict the phenological changes of the predicted image (such as amplified areas). Compared with other methods, the proposed method obtains the best effect in recovering phenological changes and achieves high-fidelity spectral performance. In addition to qualitative experiments, quantitative evaluations further substantiate the superior performance of the proposed CIG-STF model among various fusion methods. Tables I and II display the quantitative experimental results of phenological changes. The outcomes of the quantitative experiment provide additional validation for our inferences drawn from the qualitative experimental results; our proposed method outperforms other comparison methods in every aspect.

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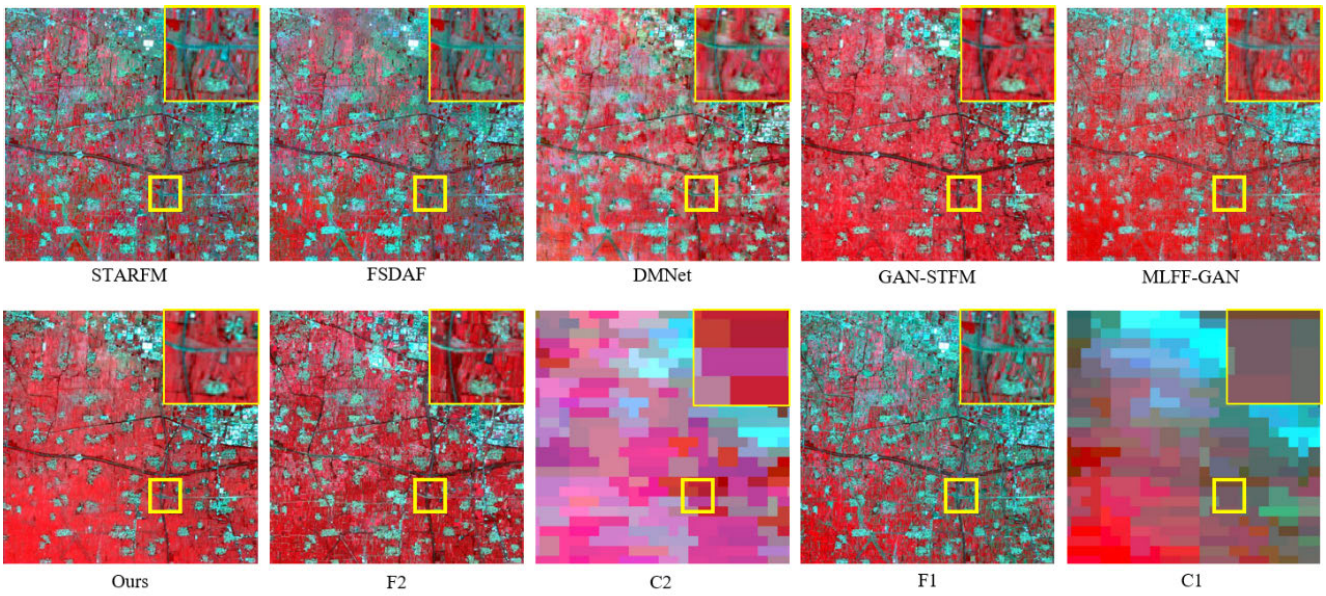


Fig. 10. Fusion results for images with land cover changes in the Daxing dataset. Among them, F1 and C1 are the fine and coarse images, respectively; at the auxiliary moment May 18, 2018, C2 is the coarse image at the target moment September 12, 2017; and F2 is the ground truth.

TABLE I

QUANTITATIVE EVALUATION RESULTS WITH PHENOLOGICAL CHANGES IN THE LGC DATASET ON MARCH 2, 2005 (THE TOP-PERFORMING RESULT EMPHASIZED IN BOLD AND THE SECOND-BEST RESULT UNDERLINED)

Methods	SAM	RMSE	SSIM	CC
STARFM	0.0972	0.0175	0.9393	0.9781
FSDAF	0.0989	0.0178	0.9197	0.9774
DMNet	0.0941	0.0175	0.9425	0.9767
GAN-STFM	0.1033	0.0192	0.9284	0.9740
MLFF-GAN	0.0969	0.0187	0.9359	0.9754
Ours	**0.0919**	**0.0169**	**0.9430**	**0.9784**

TABLE II

QUANTITATIVE EVALUATION RESULTS WITH PHENOLOGICAL CHANGES IN THE DAXING DATASET ON DECEMBER 14, 2016 (THE TOP-PERFORMING RESULT EMPHASIZED IN BOLD AND THE SECOND-BEST RESULT UNDERLINED)

Methods	SAM	RMSE	SSIM	CC
STARFM	0.2304	0.0226	0.7963	0.8868
FSDAF	0.2341	0.0229	0.7908	0.8827
DMNet	0.2002	0.0204	0.8349	0.9095
GAN-STFM	0.2047	0.0205	0.8312	0.9104
MLFF-GAN	0.2087	0.0205	0.8242	0.9102
Ours	**0.2001**	**0.0203**	**0.8357**	**0.9171**

2) Experiments for Land Cover Changes: We conducted experiments on the LGC datasets and Daxing datasets, to demonstrate the superiority of our proposed CIG-STF model in areas with significant changes in land cover. Figs. 9 and 10 show the qualitative experimental results obtained by STARFM, FSDAF, DMNet, GAN-STFM, MLFF-GAN, and our proposed method on December 12, 2004 and September 12, 2017, respectively. The visualization results show that almost all comparison methods have better visual effects, but there are still some gaps. Specifically, STARFM exhibits notable spectral distortions that significantly deviate from ground truth, accompanied by pronounced object edge curvature and suboptimal water body recovery in the amplified areas. FSDAF displays evident spatial distortions and spectral aberrations, leading to blurred object edges. DMNet experiences severe spectral distortion, possibly attributable to the utilization of a limited set of convolutions in the architecture. Consequently, the model exhibits poor recovery performance in areas characterized by substantial changes in land cover. MLFF-GAN demonstrates bad efficacy in restoring the alterations present in the image at the specified target moment. Similar to the approach employed in this study, GAN-STFM relies exclusively on two images as inputs to the model. Nevertheless, spatial distortions in some regions and spectral aberrations in water bodies are evident in its outcomes. By contrast, our proposed CIG-STF method attains superior outcomes, exceling in both the recovery of spatial details and the faithfulness of spectral details. Similarly, as in the experiments on phenological changes, we conducted corresponding quantitative experiments on land cover changes, as shown in Tables III and IV. Consistent with our earlier descriptions, the quantitative metric results exhibit the same trend. Across all four metrics, our results consistently demonstrate superior effectiveness, indicating that our proposed network enhanced by change information guidance performs well in areas experiencing land cover changes.

In summary, both qualitative and quantitative experiments consistently demonstrate the competitive advantages of our proposed CIG-STF method. Therefore, our model can achieve precise fusion effects in terms of land cover changes and shows strong practical applicability.

Combining the experimental analyses of land cover changes and phenological variations, we can conclude that the proposed CIG-STF method achieves optimal spatial structure reconstruction and spectral detail restoration in regions with significant land cover and phenological changes. This further

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TABLE III

QUANTITATIVE EVALUATION RESULTS WITH LAND COVER CHANGES IN THE LGC DATASET ON DECEMBER 12, 2004 (THE TOP-PERFORMING RESULT EMPHASIZED IN BOLD AND THE SECOND-BEST RESULT UNDERLINED)

Methods	SAM	RMSE	SSIM	CC
STARFM	0.3082	0.0356	0.7614	0.7860
FSDAF	0.2961	0.0350	0.7616	0.7899
DMNet	0.2924	0.0360	0.7846	0.7958
GAN-STFM	0.2847	0.0332	0.7867	0.8244
MLFF-GAN	0.3170	0.0358	0.7716	0.7829
Ours	**0.2747**	**0.0329**	**0.8034**	**0.8306**

TABLE IV

QUANTITATIVE EVALUATION RESULTS WITH LAND COVER CHANGES IN THE DAXING DATASET ON SEPTEMBER 12, 2017 (THE TOP-PERFORMING RESULT EMPHASIZED IN BOLD AND THE SECOND-BEST RESULT UNDERLINED)

Methods	SAM	RMSE	SSIM	CC
STARFM	0.3189	0.0215	0.7938	0.9285
FSDAF	0.3201	0.0216	0.7936	0.9282
DMNet	0.2913	0.0205	0.8153	0.9385
GAN-STFM	0.2889	0.0197	0.8282	0.9407
MLFF-GAN	0.2973	0.0202	0.8155	0.9365
Ours	**0.2855**	**0.0197**	**0.8291**	**0.9417**

TABLE V

QUANTITATIVE EVALUATION RESULTS OF ABLATION EXPERIMENTS OF MDFE MODULE ON LGC DATASET AND DAXING DATASET (BEST MARKED IN BOLD)

Datasets	Prediction Date	Methods	SAM	RMSE	SSIM	CC
LGC	Mar.2th 2005	No MDFE	0.0935	0.0178	0.9417	0.9767
		Ours	**0.0919**	**0.0169**	**0.9430**	**0.9784**
	Dec.12th 2004	No MDFE	0.2787	0.0351	0.7950	0.8134
		Ours	**0.2747**	**0.0329**	**0.8034**	**0.8306**
Daxing	Dec.14th 2016	No MDFE	0.2016	0.0209	0.8347	0.9162
		Ours	**0.2001**	**0.0203**	**0.8357**	**0.9171**
	Sep.12th 2017	No MDFE	0.2877	0.0202	0.8270	0.9412
		Ours	**0.2855**	**0.0197**	**0.8291**	**0.9417**

validates the notable superiority of our approach, which enhances the generation of fused images at the predicted moment through change information guided.

D. Ablation Study

The proposed CIG-STF method comprises three main components: MDFE, STF-CD module, and reconstruction module. To validate the effectiveness of our proposed architecture, we conducted ablative experiments on each module. In addition, in order to deeply analyze the influence of the change detection task on the spatiotemporal fusion, we also conducted additional hyperparameter experiments.

The MDFE combines multiscale dilation convolution to extract feature maps from different sensor images that retain more valuable information for subsequent operations. In order to verify the effectiveness of this module, we conducted experiments using convolution with kernel 3 and Leaky ReLU to replace MDFE, called no MDFE. The quantitative results on both datasets are presented in Table V. From these tables, it is evident that removing the MDFE module leads to a decrease in the model's performance, so the MDFE module can effectively extract valuable information from the images and enhance the overall model performance.

TABLE VI

QUANTITATIVE EVALUATION RESULTS OF ABLATION EXPERIMENTS OF STF-CD MODULE ON LGC DATASET AND DAXING DATASET (BEST MARKED IN BOLD)

Datasets	Prediction Date	Methods	SAM	RMSE	SSIM	CC
LGC	Mar.2th 2005	No IAFPM	0.0947	0.0186	0.9402	0.9751
		No DFEM/CIGEM	0.0966	0.0177	0.9426	0.9778
		No CIGEM	0.0961	0.0176	0.9422	0.9780
		Ours	**0.0919**	**0.0169**	**0.9430**	**0.9784**
	Dec.12th 2004	No IAFPM	0.2832	0.0338	0.7946	0.8301
		No DFEM/CIGEM	0.2813	0.0341	0.7968	0.8116
		No CIGEM	0.2749	0.0342	0.7931	0.8175
		Ours	**0.2747**	**0.0329**	**0.8034**	**0.8306**
Daxing	Dec.14th 2016	No MDFE	0.2018	0.0206	0.8356	0.9094
		No DFEM/CIGEM	0.2014	0.0212	0.8353	0.9043
		No CIGEM	0.2042	0.0213	0.8346	0.9087
		Ours	**0.2001**	**0.0203**	**0.8357**	**0.9171**
	Sep.12th 2017	No MDFE	0.2864	0.0198	0.8276	0.9416
		No DFEM/CIGEM	0.2906	0.0209	0.8260	0.9372
		No CIGEM	0.2911	0.0210	0.8235	0.9380
		Ours	**0.2855**	**0.0197**	**0.8291**	**0.9417**

TABLE VII

QUANTITATIVE EVALUATION RESULTS OF ABLATION EXPERIMENTS OF DECAY PARAMETER ON LGC DATASET AND DAXING DATASET (BEST MARKED IN BOLD)

Datasets	Prediction Date	Methods	SAM	RMSE	SSIM	CC
LGC	Mar.2th 2005	No Decay Parameter	0.0943	0.0178	0.9391	0.9767
		Ours	**0.0919**	**0.0169**	**0.9430**	**0.9784**
	Dec.12th 2004	No Decay Parameter	0.2861	0.0344	0.7957	0.8211
		Ours	**0.2747**	**0.0329**	**0.8034**	**0.8306**
Daxing	Dec.14th 2016	No Decay Parameter	0.2025	0.0203	0.8343	0.9145
		Ours	**0.2001**	**0.0203**	**0.8357**	**0.9171**
	Sep.12th 2017	No Decay Parameter	0.2875	0.0200	0.8273	0.9414
		Ours	**0.2855**	**0.0197**	**0.8291**	**0.9417**

In addition, we introduced three submodules, i.e., IAFM, DFEM, and CIGEM, within the STF-CD module, serving as pivotal components in the network implementation. To ascertain the critical role of each module, we conducted ablation experiments on each one individually. IAFM plays a crucial role in the fusion of dual-temporal images, and the accuracy of the fused image at the target moment is pivotal for subsequent change information extraction and enhancement. Therefore, we replaced the IAFM module with a simple combination of concatenation and convolution with kernel 3 and leaky ReLU. The remaining components remained unchanged to generate the fine image at the target moment, referred to as no IAFM. To validate the effectiveness of the designed change information extraction and enhancement modules, we discarded

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TABLE VIII
QUANTITATIVE EVALUATION RESULTS OF ABLATION EXPERIMENTS OF HYPERPARAMETER A ON LGC AND DAXING DATASET (BEST MARKED IN BOLD)

A	Mar.2th 2005				Dec.12th 2004				Dec.14th 2016				Sep.12th 2017			
	SAM	RMSE	SSIM	CC	SAM	RMSE	SSIM	CC	SAM	RMSE	SSIM	CC	SAM	RMSE	SSIM	CC
0.1	0.0942	0.0173	0.9424	0.9783	0.2827	0.0330	.07991	0.8258	0.2010	0.0205	0.8347	0.9164	**0.2840**	0.0198	0.8305	0.9413
0.2	0.0918	0.0199	0.9371	0.9781	0.2830	0.0334	0.7929	0.8127	**0.1990**	0.0204	0.8351	0.9142	0.2892	0.0199	0.8243	0.9408
0.3	0.0919	**0.0169**	**0.9430**	0.9784	**0.2747**	0.0329	**0.8034**	**0.8306**	0.2001	0.0203	**0.8357**	**0.9171**	0.2855	**0.0197**	**0.8291**	**0.9417**
0.4	0.0924	0.0174	0.9383	0.9775	0.2833	0.0337	0.7950	0.8252	0.2046	0.0209	0.8256	0.9122	0.2935	0.0200	0.8197	0.9398
0.5	0.0929	0.0210	0.9355	**0.9788**	0.2747	**0.0321**	0.7987	0.8240	0.2010	0.0210	0.8338	0.9137	0.2927	0.0205	0.8238	0.9396
0.6	0.0922	0.0170	0.9419	0.9786	0.2825	0.0348	0.7903	0.8085	0.2056	0.0202	0.8334	0.9163	0.2887	0.0199	0.8249	0.9415
0.7	0.0935	0.0174	0.9414	0.9783	0.2822	0.0334	0.7955	0.8174	0.2015	0.0209	0.8314	0.9113	0.2909	0.0203	0.8235	0.9413
0.8	**0.0914**	0.0174	0.9422	0.9785	0.2835	0.0331	0.7895	0.8177	0.2009	**0.0201**	0.8328	0.9149	0.2896	0.0197	0.8225	0.9407
0.9	0.0932	0.0180	0.9386	0.9779	0.2809	0.0330	0.7959	0.8213	0.2085	0.0212	0.8307	0.9085	0.2905	0.0203	0.8266	0.9393
1.0	0.0929	0.0180	0.9386	0.9779	0.2887	0.0337	0.7892	0.8049	0.2036	0.0202	0.8332	0.9147	0.2920	0.0202	0.8250	0.9412

DFEM and CIGEM, focusing solely on the reconstruction of the target moment image, referred to as no DFEM/CIGEM. To confirm that the change information extracted by DFEM can guide the generation of a more accurate target moment image, we excluded CIGEM and used only IAFM and DFEM to generate the prediction image, termed no CIGEM. Quantitative results of the ablative experiments on the three modules conducted on the LGC and Daxing datasets are presented in Table VI. From these tables, it is evident that the proposed IAFM, DFEM, and CIGEM modules contribute to enhancing the model performance, resulting in the generation of more satisfactory high-fidelity prediction results.

Finally, to validate the effectiveness of the designed dynamic decay loss for capturing more realistic change information, we removed the decay parameter (19), referred to as no decay parameter. The quantitative results on the LGC and Daxing datasets are presented in Table VII. It can be observed that the dynamic loss initially strengthens the constraint on change information, gradually reducing the weight of change information later on and focusing on the generation of the fusion results. This strategic approach leads to improving the network's performance and the generation of more accurate fusion results.

In addition, in order to verify the impact of change detection during network training, we conducted the ablation experiments on hyperparameter A in (19). The experimental results are meticulously presented in Table VIII. By analyzing this table, it can be revealed that setting A to 0.3, both LGC and Daxing datasets win substantial benefits. This can demonstrate that when hyperparameter A is set to 0.3, change detection is most effective at assisting spatiotemporal fusion.

V. CONCLUSION

In this article, we have introduced a novel spatiotemporal fusion method named CIG-STF, aiming to alleviate the shortcomings of existing approaches in scenarios with substantial land cover changes. We designed an MDFE, an STF-CD, and an image reconstruction module. Specifically, the feature maps obtained through the MDFE are input into the STF-CD module, which consists of three submodules with distinct roles. IAFM generates preliminary fusion feature maps through attention strategies, DFEM extracts the difference maps between the preliminary fusion feature maps and the feature maps of the previous moment images, and CIGEM guides the difference maps to enhance the preliminary fusion results to obtain the final prediction results. Finally, we designed a reconstruction module and a dynamic hybrid loss function to constrain and enhance spatial structure and spectral details. The experimental results demonstrate that our proposed CIG-STF method outperforms state-of-the-art spatiotemporal fusion methods in both land cover changes and phenological changes.

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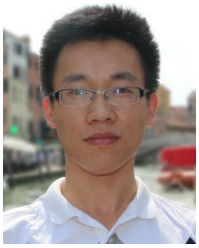
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